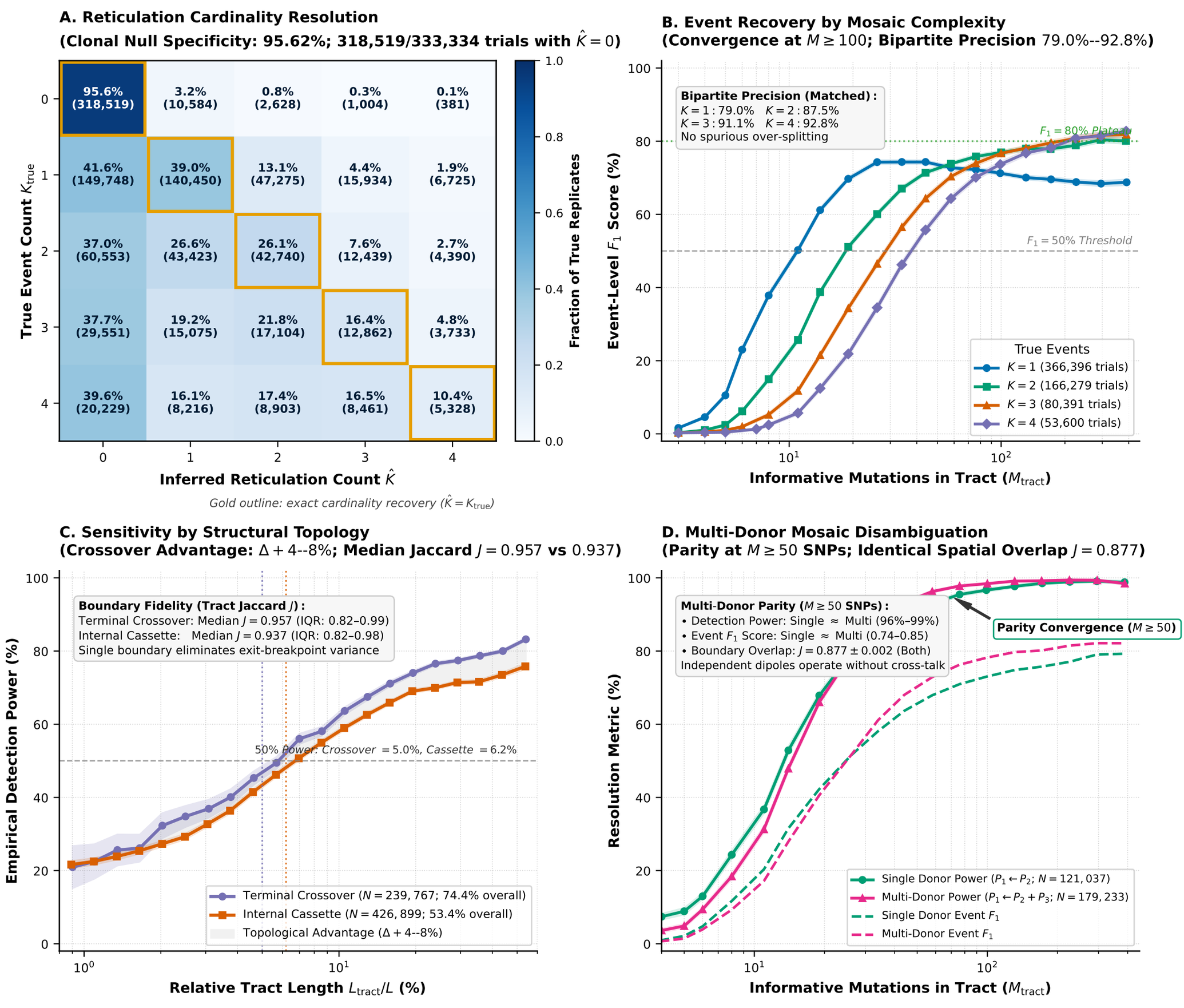

A. Reticulation Cardinality Resolution
(Clonal Null Specificity: 95.62%; 318,519/333,334 trials with $\hat{K} = 0$)
95.6% (318,519)
3.2% (10,584)
0.8% (2,628)
0.3% (1,004)
0.1% (381)
41.6% (149,748)
39.0% (140,450)
13.1% (47,275)
4.4% (15,934)
1.9% (6,725)
37.0% (60,553)
26.6% (43,423)
26.1% (42,740)
7.6% (12,439)
2.7% (4,390)
37.7% (29,551)
19.2% (15,075)
21.8% (17,104)
16.4% (12,862)
4.8% (3,733)
39.6% (20,229)
16.1% (8,216)
17.4% (8,903)
16.5% (8,461)
10.4% (5,328)
True Event Count K_{true}
Inferred Reticulation Count $\hat{K}$
Fraction of True Replicates
Gold outline: exact cardinality recovery ($\hat{K} = K_{true}$)
B. Event Recovery by Mosaic Complexity
(Convergence at $M \geq 100$; Bipartite Precision 79.0%--92.8%)
Bipartite Precision (Matched):
$K = 1$: 79.0% $K = 2$: 87.5%
$K = 3$: 91.1% $K = 4$: 92.8%
No spurious over-splitting
$F_1 = 80\%$ Plateau
$F_1 = 50\%$ Threshold
True Events
$K = 1$ (366,396 trials)
$K = 2$ (166,279 trials)
$K = 3$ (80,391 trials)
$K = 4$ (53,600 trials)
Event-Level F_1 Score (%)
Informative Mutations in Tract (M_{tract})
C. Sensitivity by Structural Topology
(Crossover Advantage: $\Delta + 4$--8%; Median Jaccard $J = 0.957$ vs 0.937)
Boundary Fidelity (Tract Jaccard J):
Terminal Crossover: Median $J = 0.957$ (IQR: 0.82–0.99)
Internal Cassette: Median $J = 0.937$ (IQR: 0.82–0.98)
Single boundary eliminates exit-breakpoint variance
50% Power: Crossover = 5.0%, Cassette = 6.2%
Terminal Crossover ($N = 239,767$; 74.4% overall)
Internal Cassette ($N = 426,899$; 53.4% overall)
Topological Advantage ($\Delta + 4$--8%)
Empirical Detection Power (%)
Relative Tract Length L_{tract}/L (%)
D. Multi-Donor Mosaic Disambiguation
(Parity at $M \geq 50$ SNPs; Identical Spatial Overlap $J = 0.877$)
Multi-Donor Parity ($M \geq 50$ SNPs):
• Detection Power: Single ≈ Multi (96%–99%)
• Event F_1 Score: Single ≈ Multi (0.74–0.85)
• Boundary Overlap: $J = 0.877 \pm 0.002$ (Both)
Independent dipoles operate without cross-talk
Parity Convergence ($M \geq 50$)
Single Donor Power ($P_1 \leftarrow P_2$; $N = 121,037$)
Multi-Donor Power ($P_1 \leftarrow P_2 + P_3$; $N = 179,233$)
Single Donor Event F_1
Multi-Donor Event F_1
Resolution Metric (%)
Informative Mutations in Tract (M_{tract})